2.43 Parametric Equations: Integration

Worked Solutions 1–22

1 a)

$\frac{dx}{dt} = 9t^2 - 1$ so $\int (t^2+1)(9t^2-1)\,dt$

$$= \int (9t^4 + 8t^2 - 1)\,dt = \frac{9}{5}t^5 + \frac{8}{3}t^3 - t + c$$

1 b)

$\frac{dx}{dt} = e^t$ so $\int (e^{-t}+1)e^t\,dt$

$$= \int (1 + e^t)\,dt = t + e^t + c$$

1 c)

$\frac{dx}{dt} = \ln 3 \times 3^t$ so $\int 9^t(\ln 3 \times 3^t)\,dt$

$$= \int 3^{2t}(\ln 3 \times 3^t)\,dt = \int \ln 3 \times 3^{3t}\,dt$$
$$= \tfrac{1}{3} \times 3^{3t} + c$$

Or $3^{3t-1} + c$. Or $\frac{1}{3} \times 27^t + c$.

2 a)

$$\int_1^2 (t^2+4)(6t^2+1)\,dt = \int_1^2 6t^4 + 25t^2 + 4\,dt$$
$$= \left[\tfrac{6}{5}t^5 + \tfrac{25}{3}t^3 + 4t\right]_1^2 = \left(\tfrac{1696}{15}\right) - \left(\tfrac{203}{15}\right) = \tfrac{1493}{15}$$

2 b)

$$\int_0^2 2te^t\,dt = [2te^t]_0^2 - \int_0^2 2e^t\,dt$$
$$= [2te^t - 2e^t]_0^2 = (4e^2 - 2e^2) - (0 - 2)$$
$$= 2e^2 + 2$$

2 c)

$$\int_0^1 \ln 2 \times 2^t \times 4^t\,dt = \int_0^1 \ln 2 \times 2^t \times 2^{2t}\,dt$$
$$= \int_0^1 \ln 2 \times 2^{3t}\,dt = \left[\tfrac{1}{3}2^{3t}\right]_0^1$$
$$= (\tfrac{8}{3}) - (\tfrac{1}{3}) = \tfrac{7}{3}$$

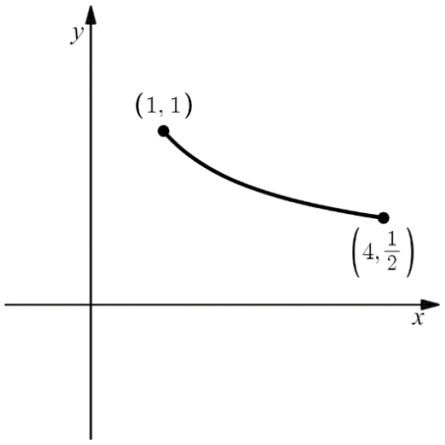
2 d)

$$\int_{0.5}^{1} -\frac{1}{t} \times t^3 \, dt = \int_{0.5}^{1} -t^2 \, dt = \left[-\tfrac{1}{3}t^3\right]_{0.5}^{1}$$
$$= (-\tfrac{1}{3}) - (-\tfrac{1}{24}) = -\tfrac{7}{24}$$

3 a)

When $t = 1$, $(1, 1)$ and when $t = 2$, $(4, \frac{1}{2})$. $t = \frac{1}{y}$ so $x = (\frac{1}{y})^2 \Rightarrow xy^2 = 1$.

3 b)



3 c)

$\int_1^2 \frac{1}{t} \times 2t \, dt = \int_1^2 2 \, dt = [2t]_1^2 = 4 - 2 = 2.$

4 a)

When $x = 4$, $4 = 2t - 3$ so $t = \frac{7}{2}$. When $x = 0$, $0 = 2t - 3$ so $t = \frac{3}{2}$.

4 b)

$$\int_{\frac{3}{2}}^{\frac{7}{2}} \frac{4}{t} \times 2 \, dt = [8 \ln t]_{\frac{3}{2}}^{\frac{7}{2}} = (8 \ln \tfrac{7}{2}) - (8 \ln \tfrac{3}{2})$$
$$= 8 \ln \tfrac{7}{3}$$

4 c)

$t = \frac{1}{2}x + \frac{3}{2}$ so $y = \frac{4}{\frac{1}{2}x + \frac{3}{2}} = \frac{8}{x+3}$.

$$\int_0^4 \frac{8}{x+3} \, dx = [8 \ln(x+3)]_0^4 = (8 \ln 7) - (8 \ln 3)$$
$$= 8 \ln \tfrac{7}{3}$$

5 a)

When $y = 0$, $2t^2 - t^3 = t^2(2 - t) = 0$ so either $t = 0$ or $t = 2$. When $t = 0$, $x = 0^2 + 2 \times 0 = 0$. When $t = 2$, $x = 2^2 + 2 \times 2 = 8$, therefore $k = 8$.

5 b)

$$\begin{aligned}&\int_0^2 (2t^2 - t^3)(2t + 2)\,dt \\ &= \int_0^2 -2t^4 + 2t^3 + 4t^2\,dt \\ &= \left[-\tfrac{2}{5}t^5 + \tfrac{1}{2}t^4 + \tfrac{4}{3}t^3\right]_0^2 = (\tfrac{88}{15}) - (0) = \tfrac{88}{15}\end{aligned}$$

6 a)

$x = 0$ so $0 = t^2(t - 1)$ so $t = 0$ or $t = 1$. P occurs when $t = 1$.

6 b)

When $x = 2$, $t^2(t - 1) = 4$. $t^3 - t^2 - 4 = 0$ which has solution $t = 2$.

$$\begin{aligned}\int_1^2 (3t^2 - 2t)t^2\,dt &= \int_1^2 3t^4 - 2t^3\,dt \\ &= \left[\tfrac{3}{5}t^5 - \tfrac{1}{2}t^4\right]_1^2 = (\tfrac{56}{5}) - (\tfrac{1}{10}) = \tfrac{111}{10}\end{aligned}$$

7 a)

When $t = 0$, $(0, \frac{1}{2})$ and when $t = 5$ so $(\frac{5}{9}, \frac{8}{81})$.

7 b)

$x = \frac{t}{t+4}$ so $\frac{dx}{dt} = \frac{(t+4)-t}{(t+4)^2} = \frac{4}{(t+4)^2}$.

$$\int_0^5 \frac{8}{(t+4)^2} \times \frac{4}{(t+4)^2}\,dt = \int_0^5 \frac{32}{(t+4)^4}\,dt \text{ as req}$$

7 c)

$\left[-\frac{32}{3(t+4)^3}\right]_0^5 = (-\frac{32}{2187}) - (-\frac{32}{192}) = \frac{665}{4374}$.

8 a)

When $x = 0$, $\tan\theta = 0$ so $\theta = 0 \Rightarrow (0, 1)$. When $x = \sqrt{3}$, $\tan\theta = \sqrt{3}$ so $\theta = \frac{\pi}{3} \Rightarrow (\sqrt{3}, \frac{1}{8})$.

8 b)

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sec^2\theta \cos^3\theta\,d\theta &= \int_0^{\frac{\pi}{3}} \cos\theta\,d\theta = [\sin\theta]_0^{\frac{\pi}{3}} \\ &= (\tfrac{\sqrt{3}}{2}) - (0) = \tfrac{\sqrt{3}}{2} \text{ as req.}\end{aligned}$$

9 a)

When $x = 0$, $0 = 3\sin t$ so $t = 0$ or $t = \frac{\pi}{2}$. $\frac{dx}{dt} = 3\cos t$. So $y\frac{dx}{dt} = 2\sin 2t\, 3\cos t = 6(2\sin t\cos t)\cos t$ which gives $\int_0^{\frac{\pi}{2}} 12\sin t\cos^2 t\, dt$ as req.

9 b)

$$\int_0^{\frac{\pi}{2}} 12\sin t(\cos t)^2\, dt = [-4(\cos t)^3]_0^{\frac{\pi}{2}}$$
$$= (0) - (-4) = 4 \text{ as req.}$$

10 a)

When $x = 0$, $t = 0$. When $x = 1$, $t = 1$. $\frac{dx}{dt} = 3t^2$. So since $\int_0^1 y\, dx = 3$,

$$\int_0^1 k\sqrt{t} \times 3t^2\, dt = 3 \Rightarrow k\int_0^1 3t^{\frac{5}{2}}\, dt = 3 \text{ as req.}$$

10 b)

$k[\frac{6}{7}t^{\frac{7}{2}}]_0^1 = 3 \Rightarrow k[(\frac{6}{7}) - (0)] = 3$, therefore $k = \frac{7}{2}$.

11 a)

When $t = 0$, $(0, \frac{1}{2})$ and when $t = \pi$, $(\pi, \frac{1}{2})$.

11 b)

$x = t + \sin t$ so $\frac{dx}{dt} = 1 + \cos t$.

$$\int_0^{\pi} (\tfrac{1}{2}\cos^2 t) \times (1 + \cos t)\, dt$$
$$= \tfrac{1}{2}\int_0^{\pi} (1 - \sin^2 t)(1 + \cos t)\, dt$$
$$= \tfrac{1}{2}\int_0^{\pi} (1 - \sin^2 t + \cos t - \cos t\sin^2 t)\, dt$$

11 c)

$$\tfrac{1}{2}\int_0^{\pi} (1 - \sin^2 t + \cos t - \cos t\sin^2 t)\, dt$$
$$= \tfrac{1}{2}\int_0^{\pi} (1 - \tfrac{1}{2} - \tfrac{1}{2}\cos 2t + \cos t - \cos t\sin^2 t)\, dt$$
$$= \tfrac{1}{2}\int_0^{\pi} (\tfrac{1}{2} - \tfrac{1}{2}\cos 2t + \cos t - \cos t\sin^2 t)\, dt$$
$$= \tfrac{1}{2}[\tfrac{1}{2}t - \tfrac{1}{4}\sin 2t + \sin t - \tfrac{1}{3}\sin^3 t]_0^{\pi}$$
$$= \tfrac{1}{2}[(\tfrac{1}{2}\pi) - (0)] = \tfrac{1}{4}\pi$$

12 a)

$\int x\,dy$ is required: $x = t^2$ and $\frac{dy}{dt} = -\frac{6}{t^2}$. When $y = 0$, $t = 3$. When $y = 5$, $t = \frac{6}{7}$.

$$\int_3^{\frac{6}{7}} t^2 \times -\frac{6}{t^2}\,dt = \int_3^{\frac{6}{7}} -6\,dt = \int_{\frac{6}{7}}^3 6\,dt \text{ as req}$$

12 b)

$\int_{\frac{6}{7}}^3 6\,dt = [6t]_{\frac{6}{7}}^3 = (6 \times 3) - (6 \times \frac{6}{7}) = \frac{90}{7}$.

13 a)

$\int x\,dy$ is required: $x = 5t^3$ and $\frac{dy}{dt} = -\frac{1}{t^2}$. When $y = 0$, $t = 1$. When $y = 2$, $t = \frac{1}{3}$.

$$\int_1^{\frac{1}{3}} 5t^3 \times -\frac{1}{t^2}\,dt = -\int_{\frac{1}{3}}^1 5t\,dt \text{ as req}$$

13 b)

$\int_{\frac{1}{3}}^1 5t\,dt = [\frac{5}{2}t^2]_{\frac{1}{3}}^1 = (\frac{5}{2}) - (\frac{5}{18}) = \frac{20}{9}$.

14 a)

$\frac{dx}{dt} = -\frac{12}{t^3}$.

$$\frac{dy}{dt} = -t^3e^{-t} + 3t^2e^{-t} = t^2e^{-t}(3-t)$$

$$\text{So } \frac{dy}{dx} = \frac{t^2e^{-t}(3-t)}{-\frac{12}{t^3}} = -\frac{t^5e^{-t}(3-t)}{12}$$

Max when $\frac{dy}{dx} = 0$: $t^5e^{-t}(3-t) = 0$ so $t = 0$ or $t = 3$. When $t = 3$, so $P(\frac{2}{3}, 27e^{-3})$.

14 b)

When $x = 6$, $\frac{6}{t} = 6$ so $t = 1$.

$$y\frac{dx}{dt} = (t^3e^{-t}) \times -\frac{12}{t^3} = -12e^{-t}$$

$$\begin{aligned}\text{So } \int_3^1 -12e^{-t}\,dt &= [12e^{-t}]_3^1 \\ &= (12e^{-1}) - (12e^{-3}) \\ &= \frac{12}{e} - \frac{12}{e^3} = \frac{12}{e}(1 - \frac{1}{e^2}) \text{ as req.}\end{aligned}$$

15)

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta &= \int_0^{\frac{\pi}{2}} \tfrac{1}{2} + \tfrac{1}{2}\cos(2\theta)\,d\theta \\ &= [\tfrac{1}{2}\theta + \tfrac{1}{4}\sin(2\theta)]_0^{\frac{\pi}{2}} = (\tfrac{\pi}{4} + 0) - (0 + 0) = \tfrac{\pi}{4}\end{aligned}$$

16)

When $x = 0$, $t_1 = 0$. When $x = 2$, $t_2 = \frac{\pi}{2}$.

$$\begin{aligned}&\int_0^{\frac{\pi}{2}} 4\cos t \times 2\cos t\, dt\\ &= \int_0^{\frac{\pi}{2}} 8\cos^2 t\, dt = \int_0^{\frac{\pi}{2}} 4 + 2\cos(2t)\, dt\\ &= [4t + \sin 2t]_0^{\frac{\pi}{2}} = (2\pi + 0) - (0 + 0) = 2\pi\end{aligned}$$

17 a)

When $x = 0$, $0 = 9\cos\theta$ so $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$. When $\theta = \frac{\pi}{2}$, $y = 5$, when $\theta = \frac{3\pi}{2}$, $y = -5$. $(0, 5)$ and $(0, -5)$. When $y = 0$, $0 = 5\sin\theta$ so $\theta = 0, \pi$. When $\theta = 0$, $x = 9$, when $\theta = \pi$, $y = -9$. $(9, 0)$ and $(-9, 0)$. At $A(9, 0)$, $\theta = 0$. At $B(0, 5)$, $\theta = \frac{\pi}{2}$. At $C(-9, 0)$, $\theta = \pi$.

17 b)

From $\theta = 0$ to $\theta = \pi$,

$$\frac{dx}{d\theta} = -9\sin\theta \text{ so } y\frac{dx}{d\theta} = 5\sin\theta \times -9\sin\theta$$

which gives $\int_0^{\pi} -45\sin^2\theta\, d\theta$ as req.

17 c)

$$\begin{aligned}\int_0^{\pi} -45\sin^2\theta\, d\theta &= \int_0^{\pi} -45(\tfrac{1}{2} - \tfrac{1}{2}\cos 2\theta)\, d\theta\\ &= \int_0^{\pi} -\tfrac{45}{2} + \tfrac{45}{2}\cos 2\theta\, d\theta\\ &= [-\tfrac{45}{2}t + \tfrac{45}{4}\sin 2\theta]_0^{\pi} = (-\tfrac{45}{2}\pi) - (0) = -\tfrac{45}{2}\pi\end{aligned}$$

Which is an area is $\frac{45}{2}\pi$ as req.

18 a)

When $x = \ln 2$, $\ln 2 = \ln(t - 2)$ so $t = 4$. When $x = \ln 6$, $\ln 6 = \ln(t - 2)$ so $t = 8$.

$$\frac{dx}{dt} = \frac{1}{t-2} \text{ so } y\frac{dx}{dt} = \frac{6}{t}\left(\frac{1}{t-2}\right) = \frac{6}{t(t-2)}$$

So $\int_4^8 \frac{6}{t(t-2)}\, dt$ as req.

18 b)

$\frac{6}{t(t-2)} \equiv \frac{A}{t} + \frac{B}{t-2}$. So $6 = A(t - 2) + Bt$. When $t = 0$, $6 = -2A$, when $t = 2$, $6 = 2B$. Therefore $\frac{6}{t(t-2)} = -\frac{3}{t} + \frac{3}{t-2}$.

18 c)

$$\begin{aligned}\int_4^8 \frac{6}{t(t-2)}\, dt &= \int_4^8 -\frac{3}{t} + \frac{3}{t-2}\, dt\\ &= [-3\ln t + 3\ln(t-2)]_4^8\\ &= (-3\ln 8 + 3\ln 6) - (-3\ln 4 + 3\ln 2)\\ &= 3[-\ln 8 + \ln 6 + \ln 4 - \ln 2] = 3\ln\tfrac{3}{2} = \ln\tfrac{27}{8}\end{aligned}$$

19 a)

When $x = 0$, $\frac{2t^2}{2+t} = 0$ so $t = 0 \Rightarrow (0, \frac{1}{4})$. When $x = \frac{18}{5}$, $\frac{2t^2}{2+t} = \frac{18}{5} \Rightarrow 10t^2 = 36 + 18t$

$$\Rightarrow 10t^2 - 18t - 36 = 0 \Rightarrow t = 3,\ t = -\frac{6}{5}$$

so $t = 3 \Rightarrow (\frac{18}{5}, \frac{1}{10})$.

19 b)

So from $t = 0$ to $t = 3$,

$$\frac{dx}{dt} = \frac{4t(2+t) - 2t^2}{(2+t)^2} = \frac{8t + 2t^2}{(2+t)^2} = \frac{2t(4+t)}{(2+t)^2}$$

$$\text{so } y\frac{dx}{dt} = \frac{1}{2(2+t)} \times \frac{2t(4+t)}{(2+t)^2} = \frac{t(4+t)}{(2+t)^3}$$

$\int_0^3 \frac{t(4+t)}{(2+t)^3}\,dt$ as req.

19 c)

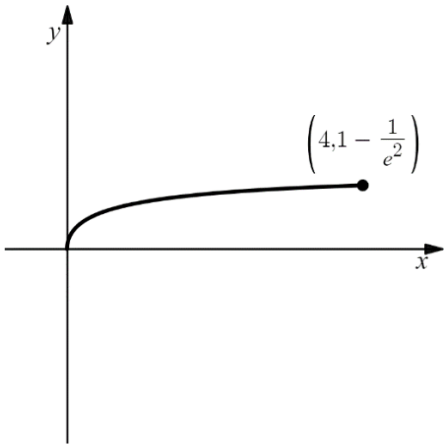
$u = 2 + t \Rightarrow \frac{du}{dt} = 1$ and $u + 2 = 4 + t$. When $t = 0$, $u = 2$, when $t = 3$, $u = 5$.

$$\begin{aligned}\int_2^5 \frac{(u-2)(u+2)}{u^3}\,dt &= \int_2^5 \frac{u^2 - 4}{u^3}\,du \\ &= \int_2^5 u^{-1} - 4u^{-3}\,du = [\ln u + 2u^{-2}]_2^5 \\ &= (\ln 5 + \tfrac{2}{25}) - (\ln 2 + \tfrac{1}{2}) = \ln(\tfrac{5}{2}) - \tfrac{21}{50}\end{aligned}$$

20 a)

When $x = 0$, $t = 0$, $y = 0$. When $x = 4$, $t = 2$, $y = 1 - e^{-2}$.

20 b)



20 c)

$\int_0^2 2t(1 - e^{-t})\,dt = \int_0^2 (2t - 2te^{-t})\,dt$ as req.

20 d)

$u = -2t$, $\frac{du}{dt} = -2$, $\frac{dv}{dt} = e^{-t}$, $v = -e^{-t}$.

$$\begin{aligned}\int_0^2 2t - 2te^{-t}\,dt &= [t^2 + 2e^{-t}]_0^2 - \int_0^2 2e^{-t}\,dt \\ &= [t^2 + 2te^{-t} + 2e^{-t}]_0^2 \\ &= (4 + 4e^{-2} + 2e^{-2}) - (0 + 0 + 2) = 2 + 6e^{-2} \text{ as req.}\end{aligned}$$

21 a)

$\int x\,dy$ is required: $x = \frac{t}{t+1}$ and $\frac{dy}{dt} = -2t$. When $y = 0$, $t = 1$. When $y = \frac{1}{2}$, $t = \frac{\sqrt{2}}{2}$.

$$\int_1^{\frac{\sqrt{2}}{2}} \frac{t}{t+1} \times -2t\,dt = \int_{\frac{\sqrt{2}}{2}}^1 \frac{2t^2}{t+1}\,dt \text{ as req.}$$

21 b)

Let $u = t + 1$ so $u - 1 = t$. $\Rightarrow 2(u-1)^2 = 2u^2 - 4u + 2$. $u = \frac{2+\sqrt{2}}{2}$ and $u = 2$.

$$\begin{aligned}&\int_{\frac{2+\sqrt{2}}{2}}^2 \frac{2u^2 - 4u + 2}{u}\,du = \int_{\frac{2+\sqrt{2}}{2}}^2 (2u - 4 + \tfrac{2}{u})\,du \\ &= [u^2 - 4u + 2\ln u]_{\frac{2+\sqrt{2}}{2}}^2 \\ &= (4 - 8 + 2\ln 2) - \left(\tfrac{3+2\sqrt{2}}{2} - 4 - 2\sqrt{2} + 2\ln(\tfrac{2+\sqrt{2}}{2})\right) \\ &= -4 + 2\ln 2 + \tfrac{5}{2} + \sqrt{2} - 2\ln(\tfrac{2+\sqrt{2}}{2}) \\ &= -\tfrac{3}{2} + \sqrt{2} + 2\ln 2 - 2\ln(\tfrac{2+\sqrt{2}}{2}) \\ &= -\tfrac{3}{2} + \sqrt{2} + 2\ln 2 - 2\ln\left(\frac{2}{\frac{2+\sqrt{2}}{2}}\right) \\ &= -\tfrac{3}{2} + \sqrt{2} + 2\ln 2 - 2\ln(4 - 2\sqrt{2}) \text{ as req.}\end{aligned}$$

Alternative Method:

$$\int \frac{2u^2 - 4u + 2}{u}\,du = \int (2u - 4 + \tfrac{2}{u})\,du = u^2 - 4u + 2\ln u$$

$(t+1)^2 - 4(t+1) + 2\ln(t+1)$. Subbing the limits in:

$$(1+1)^2 - 4(1+1) + 2\ln(1+1) = -4 + 2\ln 2$$

$$\begin{aligned}&(\tfrac{\sqrt{2}}{2} + 1)^2 - 4(\tfrac{\sqrt{2}}{2} + 1) + 2\ln(\tfrac{\sqrt{2}}{2} + 1) \\ &= \tfrac{3}{2} + \sqrt{2} - 2\sqrt{2} - 4 + 2\ln(\tfrac{\sqrt{2}}{2} + 1) \\ &= -\tfrac{5}{2} - \sqrt{2} + 2\ln(\tfrac{\sqrt{2}}{2} + 1)\end{aligned}$$

Upper limit – lower limit:

$$\begin{aligned}&-4 + 2\ln 2 - \left(-\tfrac{5}{2} - \sqrt{2} + 2\ln(\tfrac{\sqrt{2}}{2} + 1)\right) \\ &= -4 + 2\ln 2 + \tfrac{5}{2} + \sqrt{2} - 2\ln(\tfrac{\sqrt{2}}{2} + 1) \\ &= -\tfrac{3}{2} + \sqrt{2} + 2\ln\left(\frac{2}{\frac{\sqrt{2}}{2} + 1}\right) \\ &= -\tfrac{3}{2} + \sqrt{2} + 2\ln(4 - 2\sqrt{2}) \text{ as req.}\end{aligned}$$

22)

When $x = 0$, $t = 0$. When $x = 12$, $t = 3$. $\frac{dx}{dt} = 1 + 2t$. $\int_0^3 k \sin t(1 + 2t)\, dt$.

$$\begin{aligned}\int \sin t(1 + 2t)\, dt &= -(1 + 2t)\cos t + \int 2\cos t\, dt \\ &= -(1 + 2t)\cos t + 2\sin t\ (+c)\end{aligned}$$

$$\begin{aligned}\therefore \int_0^3 \sin t(1 + 2t)\, dt &= [-(1 + 2t)\cos t + 2\sin t]_0^3 \\ &= (-7\cos 3 + 2\sin 3) - (-1) = 1 + 2\sin 3 - 7\cos 3\end{aligned}$$

But since $k\int_0^3 \sin t(1 + 2t)\, dt = 10$, $k(1 + 2\sin 3 - 7\cos 3) = 10$. Therefore $k = \frac{10}{1+2\sin 3-7\cos 3}$ as req.